# A Comprehensive Survey of Time Series Forecasting: Concepts, Challenges, and Future Directions

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*(Survey Paper)*

***Abstract*—Time series forecasting (TSF) is a fundamental task in diverse decision-making scenarios, such as business intelligence, energy management, healthcare, and scientific discovery. Over the past decades, TSF has been extensively studied, with methods evolving from traditional statistical models to recent data-driven deep learning approaches. However, despite this rapid progress, the TSF literature still lacks a unified problem formulation and a systematic analysis of its core entities, data characteristics, and fundamental challenges. In particular, a systematic understanding is still needed regarding how existing TSF models address key issues such as data sparsity, long-term dependency, non-stationarity, missing observations, irregular sampling, and covariate utilization. This survey provides a comprehensive and structured overview of TSF from both problem and methodology perspectives. We first identify the key entities involved in TSF and analyze their essential characteristics, such as frequency, length, missing values, and distribution shifts. We then present a general problem formulation and summarize the major challenges that shape TSF research. Based on this foundation, we propose a taxonomy that organizes representative methods from the perspectives of preprocessing and forecasting, covering traditional statistical methods, machine learning models, deep neural architectures, and emerging data-driven paradigms such as foundation models and LLM-based forecasting. Furthermore, we discuss important research trends, including transfer learning, trustworthy forecasting, standardized evaluation, and other emerging directions. Finally, we outline promising future opportunities and deployment challenges that may further advance TSF in complex real-world scenarios.**

***Index Terms*—Time series forecasting, survey, problem formulation, taxonomy, deep learning, transfer learning.**

## I. Introduction

TIME series forecasting (TSF) aims to model the evolution of temporal data and infer future trends, providing critical support for decision-making across diverse domains, including business intelligence [1], energy management [2], healthcare [3], and scientific discovery [4]. With the rapid growth of temporal data in real-world systems, accurately predicting future observations from historical time series has become increasingly important [5]. Over the past decades, TSF methods have evolved from traditional statistical models to machine learning and deep learning approaches. However, the rapidly expanding literature still lacks a unified understanding of TSF settings, core entities, data characteristics, methodological landscape, and fundamental challenges.

Classical statistical models, such as Autoregressive Integrated Moving Average (ARIMA) [6] and Exponential Smoothing methods [7], lay the foundation for TSF. However, with the growing scale and heterogeneity of real-world time series, these methods often struggle to capture complex underlying patterns. To address this limitation, machine learning (ML) [8] and deep learning (DL) [9] introduce a data-driven paradigm that learns temporal patterns directly from data and often achieves improved forecasting performance. Nevertheless, ML- and DL-based methods still face challenges in modeling long-term dependencies, multivariate interactions, and exogenous variables. These challenges become more pronounced when covariates are partially available or observations are irregularly sampled. Moreover, real-world time series are frequently affected by noise, irregularity, and non-stationarity, making preprocessing and feature extraction essential. Therefore, techniques such as imputation [10], denoising [11], and normalization [12] are widely adopted to improve data quality and forecasting reliability.

Table I further contrasts representative surveys according to their scope and analytical perspective. Earlier surveys provide valuable foundations for understanding deep forecasting models and practical learning pipelines [8], [9]. At the same time, recent works increasingly focus on specialized architectures [13], [14], learning paradigms and benchmarks [15], [16], and emerging foundation-model or LLM-based approaches [17], [18], raising the need to clarify their relation to existing data-driven forecasting paradigms. These studies collectively map important parts of the time series analysis landscape. Still, their discussions are often organized around particular model families, emerging techniques, or language-model adaptation strategies.

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The related paper list is available at https://github.com/USTCAGI/Awesome-Papers-Time-Series-Forecasting.

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TABLE I
COMPARISON OF REPRESENTATIVE SURVEYS ON TIME SERIES FORECASTING AND ANALYSIS

| Representative Work | Year | Category | Perspective on TSF |
|---|---|---|---|
| Lim and Zohren [8] | 2021 | Deep learning survey | Focuses on neural forecasting, with limited coverage beyond it. |
| Benidis et al. [9] | 2022 | Deep learning tutorial | Centers on deep forecasting pipelines and model design. |
| Wen et al. [13] | 2023 | Architecture-specific | Focuses on Transformer variants and their adaptation to time series tasks. |
| Jin et al. [14] | 2024 | Architecture-specific | Reviews graph-based modeling for cross-variable relations. |
| Zhang et al. [15] | 2024 | Learning paradigm | Organizes pre-training and representation learning methods for time series. |
| Wang et al. [16] | 2026 | Benchmark survey | Provides systematic benchmarking of deep models across downstream tasks. |
| Su et al. [17] | 2024 | Foundation models | Reviews LLM adaptation for time series tasks. |
| Liang et al. [18] | 2024 | Foundation models | Surveys time series foundation models, emphasizing pre-training and transfer. |
| **This work** | **2026** | **Unified framework** | **Unifies formulations, data characteristics, preprocessing, methods, challenges, and future directions.** |

In contrast, TSF requires a broader synthesis that connects forecasting problems, data characteristics, preprocessing choices, and model evolution with long-standing challenges such as non-stationarity, uncertainty, and transferability.

To address the abovementioned issue, this paper provides a comprehensive survey of TSF, covering both classical statistical models and advanced data-driven approaches. We first introduce the fundamental concepts and characteristics of time series, and then review representative modeling techniques together with the challenges posed by real-world forecasting scenarios. We further emphasize the role of preprocessing in improving forecasting reliability, with particular attention to recent techniques such as tokenization and graph transformation. In addition, we discuss the methodological progress of TSF from statistical models and DL methods to transfer learning, foundation models, and trustworthy forecasting. By unifying diverse TSF methods, this survey clarifies the methodological landscape, links emerging recent trends, and highlights key future directions.

The remainder of this article is organized as follows. Section II establishes TSF foundations and summarizes evaluation protocols. Section III analyzes key characteristics and challenges of time series. Section IV reviews representative preprocessing strategies. Representative TSF approaches are then reviewed in next sections through the taxonomy illustrated in Fig. 4. Sections V and VI examine statistical and data-driven forecasting methods, respectively. Section VII discusses transfer learning for TSF, while Section VIII focuses on trustworthy TSF. Finally, Section IX outlines open challenges and future research directions.

## II. FOUNDATION CONCEPT DESCRIPTION

This section introduces the foundational concepts of TSF. We first describe the basic properties of time series. We then formalize the forecasting problem and clarify key concepts. Finally, we summarize common evaluation protocols to support fair model comparison.

### A. Introduction to Time Series

A time series instance is generally defined as chronologically ordered observations, where each data point is associated with a specific timestamp or sampling interval. Although observations are recorded in discrete form, the underlying real-world processes are often continuous and may vary over broad time and value domains. Time series are commonly characterized by several fundamental components:

- *Trend:* A long-term progression of increasing or decreasing values in the series over extended periods.
- *Seasonality:* Repeating patterns that occur at fixed intervals (e.g., daily, weekly, yearly), often caused by calendar effects or natural cycles.
- *Cyclicity:* Fluctuations that recur over time but not at fixed intervals, often linked to economic influences.
- *Random Variations:* Unpredictable changes in the series that are not explained by the trend, seasonality, or cyclical behavior, often treated as residual noise.

Overall, the chronological nature of time series observations often leads to temporal statistical dependence, which forms the basis of forecasting by enabling future values to be inferred from historical patterns.

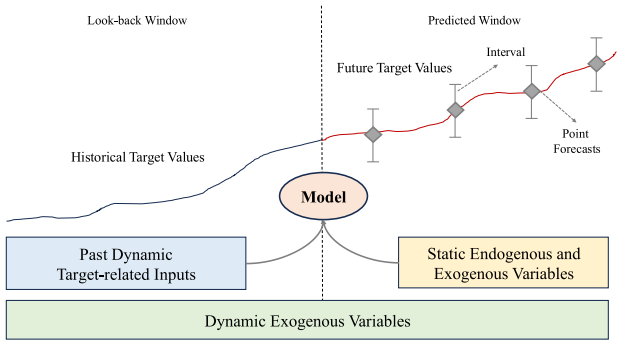



Fig. 1. Illustration of time series forecasting using historical targets and covariates to estimate future values.

### B. Basic Concepts of Time Series Forecasting

As shown in Fig. 1, time series forecasting aims to predict future values of a target series based on past observations and, potentially, additional explanatory variables. The core formulation typically involves two key components: the *look-back window* (i.e., a set of recent past observations) as input, and the *predicted window* (i.e., one or more future time steps) as output. Below, we outline several fundamental concepts:

- *Look-back Window of the Target Series:* A sequence of recent observations (e.g., the past $L$ time steps) from the

target series. It provides historical context by capturing trends, seasonality, and temporal dependencies.

- *Covariates (Exogenous Variables):* Forecasting tasks often use external factors beyond the target series, such as weather or calendar attributes. In this survey, we distinguish *known-in-advance covariates*, whose future values are available at prediction time, from *partially observed covariates*, whose future values are unavailable, uncertain, or noisy. This distinction is important for model design and evaluation, as unavailable future covariates may introduce information leakage.
- *Predicted Window of the Target Series:* The set of future time steps to be predicted. This window can vary in length, from a single step ahead (e.g., the next hour) to a longer horizon (e.g., the whole next week or month).
- *Univariate vs. Multivariate Forecasting: Univariate forecasting* uses only one target series without external inputs. *Multivariate forecasting* involves multiple related series or covariates, allowing the model to learn from interdependencies and potentially improve accuracy.
- *Iterative vs. Direct Strategies: Iterative forecasting* predicts future values step by step by recursively using previous predictions, enabling flexible horizons and autoregressive generation but risking error accumulation. *Direct forecasting* predicts each future step or the whole horizon directly, reducing recursive errors and supporting long-horizon forecasting, but often requiring horizon-specific modeling.

In summary, forecasting relies on selecting informative historical context and appropriate covariates to learn patterns that can predict future values. A clear understanding of these concepts is essential for building effective forecasting models.

### C. Problem Definition of Time Series Forecasting

Depending on the form of the output, forecasting tasks can be classified into two main categories: non-probabilistic (point-based) and probabilistic forecasting.

*1) Point-Based Time Series Forecasting:* Point-based forecasting generates a deterministic estimate for each future time step. These methods typically optimize predefined error metrics, such as mean absolute error (MAE) or root mean square error (RMSE), without explicitly modeling predictive uncertainty. Formally, given a time series $\mathbf{y} = \{y_1, y_2, \ldots, y_T\}$, the objective is to estimate future values $\hat{\mathbf{y}} = \{\hat{y}_{T+1}, \hat{y}_{T+2}, \ldots, \hat{y}_{T+\tau}\}$, where each $\hat{y}_t$ is a point estimate. Under squared error minimization, it is commonly defined as the conditional expectation:

$$\hat{y}_t = \mathbb{E}[y_t \mid \mathbf{y}].$$

Representative approaches include ARIMA, support vector regression (SVR), and neural forecasting models.

*2) Probabilistic Time Series Forecasting:* Probabilistic forecasting, in contrast, models uncertainty by predicting a full probability distribution over future values. Rather than outputting a single estimate, the model generates a conditional probability density function (PDF) $p(y_{T+1:T+\tau} \mid \mathbf{y})$. This enables richer insights, such as computing prediction intervals and quantifying risk. A typical formulation is:

$$y_t \sim p(y_t \mid \mathbf{y}),$$

where $p(y_t \mid \mathbf{y})$ can be explicitly parameterized, for example, as a Gaussian distribution $\mathcal{N}(\mu_t, \sigma_t^2)$ whose parameters are learned by the model. It can also be modeled using more advanced generative approaches, such as DeepAR [19], Variational Autoencoders [20], or flow-based models [21].

### D. Evaluation Protocols

Given the strong dependency structure and sequential nature of time series, well-designed evaluation protocols are essential for assessing forecasting performance. These protocols specify data splits, define appropriate metrics, and help prevent data leakage, thereby supporting fair, rigorous, and reproducible comparison across forecasting methods [16], [22]. As TSF tasks, datasets, and models diversify, standardized benchmarks and protocols are needed for consistent comparison. In practice, time series are typically divided into training, validation, and test sets, with ratios such as 7:2:1 depending on the dataset scale and experimental setting. Unlike standard i.i.d. tasks, such splits should preserve chronological order to avoid future information leakage. Reliable evaluation also requires avoiding common pitfalls, such as random splitting, leakage from future covariates or normalization statistics, and unfair comparisons caused by inconsistent look-back windows, forecasting horizons, preprocessing pipelines, or tuning budgets. Accordingly, preprocessing parameters should be estimated only from the training set, and model comparisons should adopt consistent input lengths, horizons, metrics, and validation protocols.

Table II summarizes the formulas, advantages, and limitations of commonly used evaluation metrics. These metrics can be broadly categorized into *point-based* and *probabilistic* metrics. Point-based metrics measure the accuracy of single-valued forecasts, while probabilistic metrics evaluate the quality of predictive distributions. For short-term point forecasting, MAE and RMSE are commonly used to measure local prediction errors. For long-horizon forecasting, horizon-wise MAE or MSE can better reveal how errors evolve across prediction steps. For probabilistic forecasting, CRPS, quantile loss, and NLL are more suitable because they assess predictive distributions rather than single-point estimates.

## III. Challenges Analysis

This section analyzes the major challenges in TSF. We discuss data quality issues, irregular sampling, long-term dependency modeling, multivariate interactions, exogenous variable integration, distribution shifts, multi-scale representations, and model generalizability.

### A. Noise, Anomalies, and Outliers

Time series, often obtained from multiple sensors or measurement processes, inevitably raise concerns regarding data quality. In practice, they may contain noise, anomalies, and outliers, which can undermine the performance of downstream

TABLE II
OVERVIEW OF WIDELY USED EVALUATION METRICS FOR POINT-BASED AND PROBABILISTIC FORECASTING TASKS

| Task | Metric | Formula | Strengths | Limitations |
|---|---|---|---|---|
| Point-based Forecasting | MAE | $\frac{1}{n}\sum_{t=1}^{n}\lvert y_t-\hat{y}_t\rvert$ | Interpretable and less affected by occasional large errors. | Treats all deviations linearly and may underemphasize large errors. |
| | MSE | $\frac{1}{n}\sum_{t=1}^{n}(y_t-\hat{y}_t)^2$ | Strongly penalizes large deviations and is optimization-friendly. | Sensitive to outliers and scale-dependent. |
| | SMAPE | $\frac{1}{n}\sum_{t=1}^{n}\frac{2\lvert y_t-\hat{y}_t\rvert}{\lvert y_t\rvert+\lvert\hat{y}_t\rvert}$ | Scale-normalized and suitable for comparing across series. | Unstable when both true and predicted values are close to zero. |
| | MAPE | $\frac{1}{n}\sum_{t=1}^{n}\left\lvert\frac{y_t-\hat{y}_t}{y_t}\right\rvert$ | Intuitive percentage error for business-oriented scenarios. | Undefined or inflated when ground-truth values are zero or very small. |
| | MASE | $\frac{\frac{1}{n}\sum_{t=1}^{n}\lvert y_t-\hat{y}_t\rvert}{\frac{1}{T-1}\sum_{t=2}^{T}\lvert y_t^{\mathrm{tr}}-y_{t-1}^{\mathrm{tr}}\rvert}$ | Scale-free and comparable across datasets. | Depends on the selected naive baseline and seasonal assumptions. |
| | OWA | $\frac{1}{2}\left(\mathrm{SMAPE}_r+\mathrm{MASE}_r\right)$ | Combines relative SMAPE and MASE performance. | Depends on the reference baseline and aggregation protocol. |
| Probabilistic Forecasting | CRPS | $\int_{-\infty}^{\infty}\left(F(z)-\mathbb{I}\{z\geq y\}\right)^2 dz$ | Evaluates the full predictive distribution with intuitive units. | Mainly defined for univariate targets and can be costly to estimate. |
| | $\rho$-QL | $2(\rho-\mathbb{I}\{y<\hat{q}_\rho\})(y-\hat{q}_\rho)$ | Evaluates quantile calibration at risk levels. | Requires quantile selection. |
| | NLL | $-\log p_\theta(y\mid x)$ | Principled probabilistic scoring rule. | Sensitive to misspecification and less intuitive. |
| | VS | $\sum_{i,j} w_{ij}\left(\Delta_{ij}^{p}-\hat{\Delta}_{ij}^{p}\right)^2$ | Captures dependence structure in multivariate probabilistic forecasts. | Computationally expensive and requires pairwise weights. |

forecasting tasks. Although substantial efforts have been devoted to imputation and anomaly detection [23], [24], mainstream forecasting studies often assume that benchmark datasets are well-filtered and of high quality. Therefore, it remains challenging to develop effective methods that mitigate the impact of low-quality observations on forecasting models.

### B. Irregular Time Series

Although continuous formulations are widely adopted in TSF, many real-world time series are irregularly sampled due to technical constraints, missing observations, or limitations in the data collection process. As a result, observations may be recorded at non-uniform time intervals rather than following a fixed sampling frequency. These temporal gaps can themselves provide important information about the underlying process, such as changes in system states or observation availability. Therefore, irregular sampling requires forecasting models to handle variable intervals and evolving latent dynamics [25], [26], with TiWeaver [27] further capturing contextual temporal dynamics via patching.

### C. Long-Term Sequence Forecasting

Time series typically consist of continuous numerical values and can be viewed as an intermediary modality between image and language data, while retaining distinct sequential characteristics. Modeling temporal dependencies remains a central focus in time series forecasting [28]. From a sequence perspective, time series often span long intervals, leading to extended input sequences. From a numerical perspective, their values usually lack fixed bounds, which further increases forecasting difficulty. Many real-world applications are formulated as long-term TSF tasks [1], [3], [4], requiring models to predict over extended horizons from long historical windows. This setting makes it challenging to capture long-range dependencies and maintain stable prediction quality. Moreover, recursive multi-step forecasting is vulnerable to error accumulation during numerical sequence modeling [29]. Representative modeling strategies, including patch-wise tokenization and Transformer-style long-range dependency modeling, are further discussed in Sections IV and VI-B.

### D. Multivariate Dependence Modeling

Collected time series datasets often exhibit multivariate characteristics due to complex cross-channel dependencies. In scenarios such as traffic load prediction and temperature analysis, causal or leading effects may exist across variables [30]. However, such dependencies are difficult to identify in complex and high-dimensional systems without clear prior assumptions or experiential guidance. Meanwhile, in systems such as the human body or weather, relevant variables are difficult to predefine, requiring multi-perspective data collection for holistic analysis [4]. Therefore, sophisticated methods are needed to capture hidden patterns and dependencies across multiple time series. Existing channel-dependent [31] and channel-independent [32] solutions still require further exploration [33]. A key challenge is to model stable dependencies while avoiding spurious cross-variable correlations. Future methods should adaptively share cross-channel information.

### E. Exogenous Variables Modeling

Real-world systems are often partially observed, making forecasts based only on endogenous variables insufficient [34], [35], [36]. Exogenous variables, such as weather, policies, holidays, and macroeconomic indicators, can influence the target series in nonlinear and time-varying ways. Therefore, a key challenge lies in identifying useful exogenous variables, distinguishing known-in-advance covariates from partially observed ones, and integrating them into forecasting models. This distinction affects model design and evaluation: known-in-advance covariates can be used during forecasting, while partially observed ones require

TABLE III
MAPPING MAJOR TSF CHALLENGES TO REPRESENTATIVE SOLUTION CATEGORIES AND RELEVANT SECTIONS

| Challenge | Typical Cause | Representative Solution Categories | Related Sections |
|---|---|---|---|
| Noise, anomalies, missing values | Sensor failures, measurement noise, abnormal events, and incomplete observations | Imputation, denoising, robust learning, probabilistic or generative reconstruction | Sec. IV, Sec. VI-D, Sec. VIII |
| Irregular sampling | Non-uniform collection intervals and observation availability changes | Continuous-time modeling, interpolation, temporal encoding | Sec. IV, Sec. VI-B |
| Long-term dependency and error accumulation | Long look-back windows, extended horizons, and recursive prediction | Patch-wise tokenization, direct forecasting, efficient attention, generative modeling | Sec. IV, Sec. VI-B, Sec. VI-D |
| Multivariate dependence | Complex cross-channel correlations and changing variable interactions | Graph transformation, channel-independent modeling, attention-based correlation learning | Sec. IV, Sec. VI-B |
| Exogenous variable availability | Future covariates may be known, uncertain, missing, or noisy | Covariate-aware architectures, joint target-covariate prediction, latent factor modeling | Sec. II, Sec. IV, Sec. VI |
| Distribution shift and non-stationarity | Evolving trends, changing seasonalities, regime shifts, and domain mismatch | Standardization, stationarization, decomposition, non-stationary attention | Sec. IV, Sec. VII, Sec. VI-B |
| Transferability and generalization | Heterogeneous domains, variable semantics, and limited labeled data | Self-supervised pretraining, foundation models, LLM-based adaptation | Sec. VII, Sec. VII-B |

imputation, joint prediction, or latent modeling. Fair evaluation should clarify whether future covariates are given, predicted, or unavailable, as this directly changes task difficulty. Related strategies are discussed in Sections IV and VI. Recent studies such as GCGNet [37] and DAG [38] further highlight graph consistency and causal relations in exogenous-variable modeling.

### F. Distribution Shift Modeling

Time series typically exhibit non-stationary characteristics, meaning that the data distribution may change rapidly over time. This leads to discrepancies across time spans and ultimately hinders the generalization ability of deep learning models, as distribution shift contradicts their fundamental assumption of independent and identically distributed (I.I.D.) data [39], [40]. Furthermore, traditional methods for addressing distribution shift, such as domain adaptation and domain generalization, may not be well suited to this task, as defining a domain for time series is not straightforward [12]. As a result, developing models or frameworks that can effectively mitigate non-stationarity remains an open challenge.

### G. Multi-Scale Representations

The multi-scale characteristics of time series are primarily reflected from two perspectives: sampling granularity and hierarchical structure [41]. On the one hand, a time series consists of discrete records sampled from a continuous space. Depending on the sampling frequency, it exhibits multi-granularity characteristics: high-frequency series are often more informative but noisier, while low-frequency series provide smoother trends but fewer details. On the other hand, time series are jointly influenced by local disturbances and global trends. Consequently, selecting an appropriate modeling granularity, exploiting multi-scale features, and integrating information across scales remain key research challenges [42], [43].

### H. Generalizability and Transferability

Unlike pixels in computer vision (CV) and word tokens in natural language processing (NLP), time series analysis lacks a standardized definition of unified semantic units. Although time series share a similar data format, their values may carry inconsistent meanings across contexts due to scenario-specific physical significance. As a result, most forecasting methodologies are typically trained and evaluated on a single dataset, limiting their generalizability and transferability. Therefore, developing a robust and scalable forecasting foundation model based on extensive multi-source pretraining data remains a significant challenge [44], [45]. Furthermore, its operational mechanisms, such as scaling laws and the influence of data distribution on model performance, remain unclear [46].

Table III summarizes how the above challenges connect to the methodological discussions in later sections. The mapping also highlights that many practical forecasting problems require combined solutions, such as coupling stationarization with domain adaptation or combining graph transformation with attention-based forecasting.

## IV. TIME SERIES PREPROCESSING

Effective preprocessing is a critical step in building robust time series forecasting models. This section reviews key techniques for addressing common data quality issues and transforming raw time series into more informative, structured, and model-friendly representations. These techniques help prepare time series for analysis and improve its suitability and reliability for downstream forecasting tasks.

### A. Time Series Imputation

Time series collected from complex systems in real-world environments are often affected by data quality. A challenge is missing data, which may arise from factors such as localized sensor failures and can lead to the loss of important temporal and cross-channel dependency information. To address this issue, time series imputation aims to restore missing values based on the observed segments. Considerable research has been devoted to this task, with existing methods generally categorized into predictive and generative approaches [10]. Predictive approaches rely on deterministic estimation techniques, including traditional neural networks such as RNNs and attention-based models [23], [47], as well as reconstruction-based learning objectives, but they often provide limited modeling of imputation uncertainty. In contrast, generative approaches employ models such as GANs and diffusion models [48], [49], enabling more

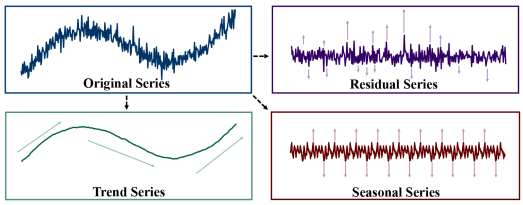



Fig. 2. An illustrative example of time series decomposition into its components: trend, seasonality, and residual.

flexible modeling of missing values and supporting uncertainty-aware imputation.

### B. Time Series Denoising

More generally, time series often suffer from significant noise contamination in addition to missing data. Notably, noise in the data not only impacts the accuracy, stability, and interpretability of model predictions but also interferes with reliable model evaluation. Consequently, denoising is an important preprocessing step for obtaining accurate, robust, and trustworthy forecasting results. Denoising approaches primarily aim to mitigate noise in time series, thereby enhancing their quality, consistency, and suitability for downstream tasks. Prominent techniques in this area include traditional decomposition-based and filtering-based methods [50], alongside more recent learning-based models [11], [51].

### C. Standardization and Stationarization

Time series often exhibit heterogeneous scales and complex distributions, making standardization and stationarization important for reliable forecasting. Although sharing a similar format, their values may carry scenario-specific physical meanings. Standardization is therefore widely used to reduce unit-induced discrepancies and stabilize model training. Stationarization methods further address non-stationarity caused by distribution shifts. DAIN [52] learns instance-wise normalization with a nonlinear network, while RevIN [12] performs reversible instance normalization by normalizing input sequences and denormalizing model outputs. Dish-TS [53] mitigates intra- and inter-space shifts through learned distribution coefficients. SAN [40] handles local distributional changes at the slice level, and SIN [54] learns normalization transformations from key statistics. These techniques improve robustness by aligning scales and mitigating distributional shifts, echoing the non-stationarity challenges in Section III-F.

### D. Time Series Decomposition Techniques

Time series decomposition is a key technique in time series analysis and forecasting. As shown in Fig. 2, the decomposition process typically involves separating the original time series into three components: the trend ($T_t$), which captures long-term changes; the seasonal component ($C_t$), which represents periodic patterns; and the residual ($S_t$), which accounts for random noise.

Decomposition is commonly formulated using additive or multiplicative models. The additive form is expressed as:

$$Y_t = T_t + C_t + S_t, \tag{1}$$

while the multiplicative form is given by:

$$Y_t = T_t \times C_t \times S_t. \tag{2}$$

By separating these components, decomposition reveals the underlying structure of time series and can improve the interpretability and effectiveness of forecasting models.

### E. Tokenization

Time series tokenization transforms raw time series into structured modeling units, referred to as *tokens*. In this survey, we distinguish between *continuous tokenization* and *discrete tokenization*. Continuous tokenization encodes time series segments into continuous embeddings, whereas discrete tokenization maps time series patterns into discrete tokens.

*1) Continuous Tokenization:* This line of methods segments raw time series into continuous modeling units, which are commonly categorized into point-wise and patch-wise approaches. Point-wise methods, such as Informer [29] and NHits [42], treat each time step as an individual unit. Although straightforward to implement, they may incur high computational costs on long sequences [55] and provide limited semantic representation for subsequent modeling [56]. Patch-wise methods address these limitations by grouping adjacent time points into fixed-length segments, which better capture local patterns. PatchTST [57] is a representative method using equal-length patches. Since then, patching strategies have been widely used in Transformers [44], [58], CNNs [59], [60], and MLPs [61]. However, fixed patching may truncate important temporal structures, such as peaks and cycles. Recent studies therefore develop more flexible patching strategies. Deformable patching [62], [63] learns patch boundaries, while multi-scale patching [64], [65] captures patterns at different resolutions. In addition, methods such as Lag-Llama [66] construct patches based on lag variables.

*2) Discrete Tokenization:* Inspired by discrete tokenization in image and audio generation [67], [68], researchers have begun representing time series as discrete tokens. Given a continuous time series $X = \{x_1, x_2, \ldots, x_T\}$ with $x_t \in \mathbb{R}$, discretization maps it to $S = \{s_1, s_2, \ldots, s_N\}$, where $s_i \in V$ and $V$ denotes a finite vocabulary. Although discretization may introduce information loss [69], it offers several advantages. Compared with continuous tokenization, discrete representations can better capture structural patterns and improve robustness to noise [70]. They also facilitate integration with other discrete modalities, such as text, thereby supporting multimodal learning [68], [71]. The most common approach in TSF is Vector Quantization Networks (VQN) [70], which learn discrete representations through deep reconstruction objectives [69], [72]. VQNs usually map time series segments into a discrete symbolic space via a quantization module, producing a compact codebook as the embedding table. They can model complex patterns and reduce sequence length [73], but require substantial training data and

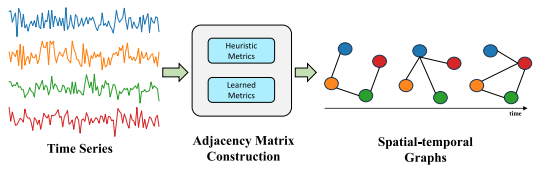



Fig. 3. Spatial-temporal graph construction for multivariate time series, where nodes denote variables and edges encode dependencies for GNN-based forecasting.

computation. Statistical discretization methods are also adopted in models such as Chronos [74] and WaveNet [71]. These methods assign values to discrete buckets according to statistical properties such as mean, variance, or quantiles. They are simple and training-free, but typically do not reduce sequence length [72].

### F. Graph Transformation

One of the most significant characteristics and challenges in multivariate time series forecasting is the complex, often non-linear correlations between variables. A viable solution is to construct a spatial-temporal graph from the original time series, where each node represents a variable and the adjacency matrix may evolve over time. These graphs are processed using Graph Neural Networks (GNNs) to effectively capture cross-channel relationships. Generally, spatial-temporal graphs can be generated through either heuristic-based or learning-based approaches [14], as illustrated in Fig. 3.

Heuristic-based methods construct spatio-temporal graphs based on the characteristics of time series. For instance, spatial distance and connectivity metrics between variables can provide guidance in typical scenarios such as traffic load forecasting [75], [76]. In addition, for forecasting tasks where prior knowledge of spatial correlations between variables is lacking, compositional methods based on data similarity metrics are widely used. These methods construct an approximate adjacency matrix based on the Pearson Correlation Coefficient (PCC) [77] or Dynamic Time Warping (DTW) distance [78] between multivariate time series. Learning-based approaches learn graph structures end-to-end, aligning graph construction with the forecasting objective and enabling implicit dependency discovery. They typically generate adjacency matrices from learned variable embeddings or parametric distributions [79], [80], [81]. Recent studies further enhance spatio-temporal forecasting by incorporating multi-view context, multi-step dependencies, gradient supervision, patch-based recurrent graph modeling, and accident-risk forecasting frameworks [82], [83], [84]. Beyond traffic forecasting, location-based temporal prediction tasks such as next point-of-interest recommendation also highlight the importance of spatial and contextual dependencies. Representative studies consider global-local context, contextualized preferences, dynamic graphs, temporal attention, contrastive learning, and disentangled personalized representations [85], [86], [87].

## V. Statistical Models

In TSF, statistical models remain important due to their solid theoretical foundations and interpretable, structured design. They leverage patterns such as lagged values and error terms to capture temporal dynamics and generate accurate forecasts, often under linear assumptions.

### A. Auto-Regressive Moving Average and Extensions

Autoregressive (AR) and moving average (MA) components are fundamental techniques in time series analysis. Several variants of the moving average (MA) model, such as the Simple Moving Average (SMA) and Exponential Moving Average (EMA), have been developed to improve accuracy. SMA smooths short-term fluctuations by averaging over a fixed window [88], while EMA assigns exponentially decreasing weights to recent observations, making it particularly responsive to trends and seasonal changes [89]. These variants enhance flexibility in capturing different patterns. The auto-regressive moving average (ARMA) model combines auto-regressive and moving-average components and is suitable for stationary time series with stable mean, variance, and autocorrelation [28]. For non-stationary data, auto-regressive integrated moving average (ARIMA) introduces differencing to make the series stationary, thereby broadening its applicability [6]. Seasonal ARIMA (SARIMA) further adds seasonal components to capture periodic patterns, but its model specification and parameter tuning are often more complex [6].

### B. Exponential Smoothing Methods

Exponential smoothing methods are widely used in time series forecasting due to their simplicity, efficiency, and adaptability. The development of these methods has focused on advances in model selection, parameter optimization, and techniques for adapting to structural changes in order to improve forecasting accuracy [90]. Simple Exponential Smoothing (SES), which assigns weights to observations using a smoothing constant $\alpha$, enables accurate predictions [91]. Several studies have compared different exponential smoothing models, highlighting the robustness and accuracy of state-space and dynamic nonlinear approaches, as well as strategies for handling uncertainties [92]. For seasonal data, the Holt-Winters method decomposes the time series into level, trend, and seasonality, supporting both additive and multiplicative forms. This makes it particularly suitable for industries with predictable seasonal demand, such as retail and energy, with recurring demand patterns [7].

### C. State-Space Models

The state-space model (SSM), derived from control theory, provides a flexible framework for modeling dynamic systems through state and observation equations. It can capture both linear and nonlinear behaviors in stationary and non-stationary time series. As a core SSM algorithm, the Kalman filter (KF) is widely used for structural and nonlinear time series modeling [93]. In addition, the hidden Markov model (HMM), a

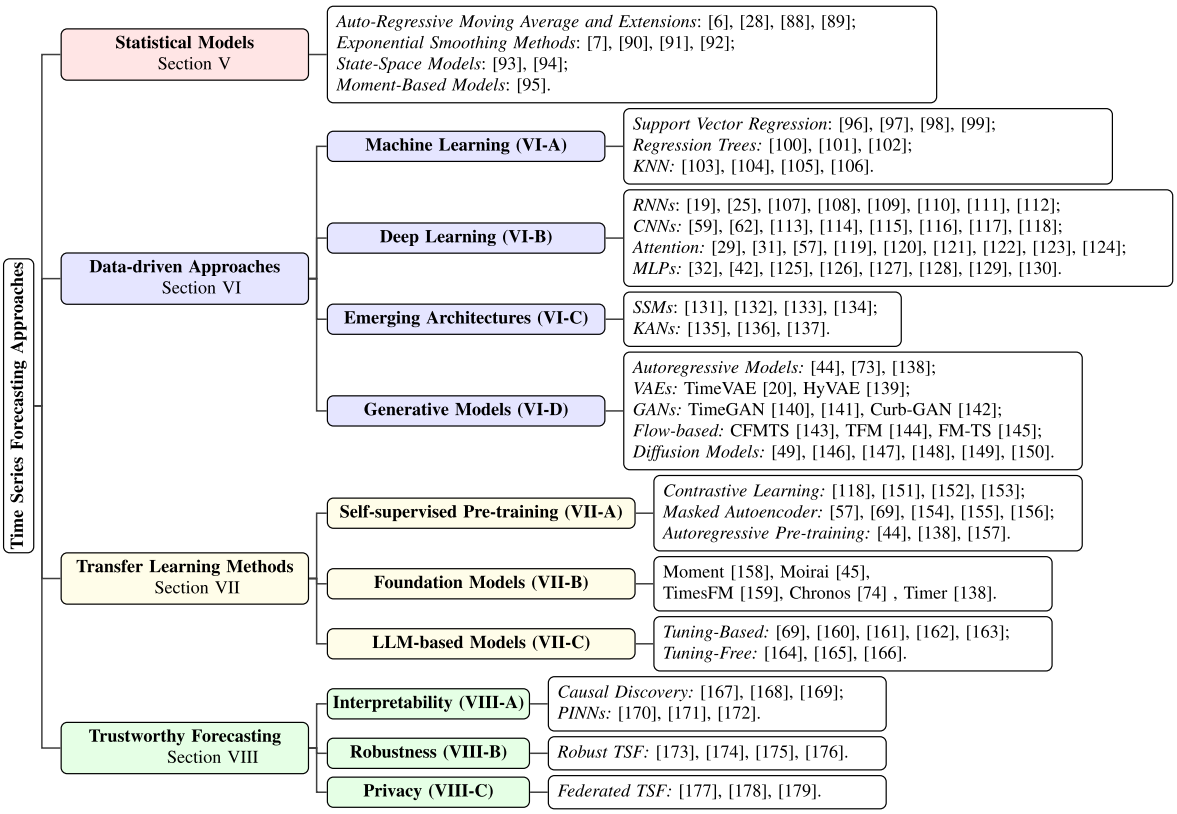



Fig. 4. Taxonomy of representative TSF approaches, organized by methodological role: statistical modeling, data-driven forecasting, transfer learning, foundation models, and trustworthy forecasting.

special case of SSM with discrete hidden states, is well suited for sequential data whose latent states evolve probabilistically. For financial time series with non-stationarity and nonlinearity, extended HMMs further incorporate exponentially weighted and dual-weighted expectation-maximization algorithms to better model complex financial dynamics and high-dimensional nonlinear patterns [94].

### D. Moment-Based Statistical Models

Moment-based statistical methods estimate model parameters by matching theoretical moments with empirical moments computed from observed time series. They are especially useful when the likelihood function is difficult to specify or optimize. For example, the Generalized Method of Moments (GMM) estimator [95] has been used to estimate Markov-switching multifractal model parameters for volatility forecasting. By relying on moment conditions rather than a fully specified likelihood, such methods can capture structural changes in financial time series while retaining a relatively interpretable statistical formulation.

*Key takeaway:* Classical statistical models remain strong baselines and practical choices when time series exhibit clear linear autocorrelation, especially in small-sample or resource-constrained settings. Their explicit assumptions make them both interpretable and efficient.

## VI. Data-Driven Approaches

In this section, we mainly cover the progress in data-driven TSF approaches, which learn the temporal patterns from the time series. We include methods ranging from machine learning methods to recent advancements in deep neural networks, with an emphasis on the latest generative models.

### A. Machine Learning Approaches

Machine learning methods are important tools for time series forecasting, offering advantages for capturing complex patterns. In this section, we will discuss three prominent approaches: Support Vector Regression (SVR), Regression Trees, and K-Nearest Neighbor (KNN) regression.

*1) Support Vector Regression:* Support Vector Regression (SVR) [180] is the regression extension of Support Vector Machines (SVM) [181], designed for continuous value prediction. Its objective is to learn a regression function $f(\mathbf{x})$ such that most data points $(\mathbf{x}_i, y_i)$ fall within an $\epsilon$-insensitive margin around the predicted values. By introducing Lagrange multipliers $\alpha_i$ and $\alpha_i^*$, SVR can be formulated in its dual form [182], leading to the following regression function:

$$f(\mathbf{x}) = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*) K(\mathbf{x}_i, \mathbf{x}) + b. \tag{3}$$

Here, $K(\mathbf{x}_i, \mathbf{x})$ denotes the kernel function, with common choices including linear, polynomial, and Gaussian radial basis function (RBF) kernels. In time series forecasting, SVR has been widely applied across various domains [183]. Representative studies use SVR for futures price movement prediction [96], air quality forecasting [97], seasonal time series modeling through Seasonal Support Vector Regression (SSVR) [98], and

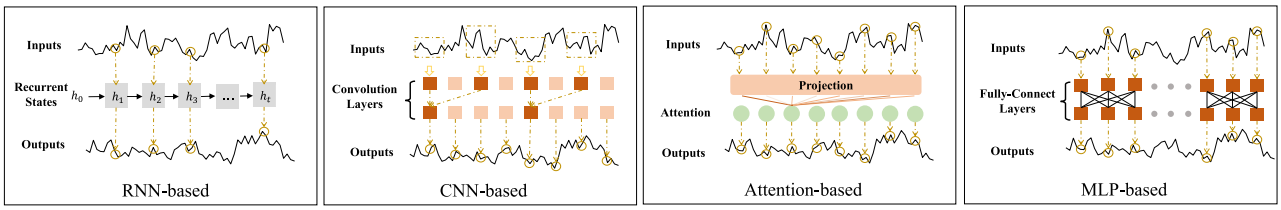



Fig. 5. Representative deep neural architectures for TSF, contrasting recurrent, convolutional, attention-based, and MLP-style designs with different temporal inductive biases.

multi-step forecasting with multiple support vector regression (MSVR) to reduce error accumulation [99].

*2) Regression Trees:* Regression trees are decision-tree-based regression methods that recursively partition the feature space into mutually exclusive regions, where each region is associated with a predicted value. Classification and Regression Trees (CART) [184] is one of the most widely used algorithms. CART recursively splits the data based on selected features and split points, with the objective of minimizing the mean squared error (MSE) within the subsets. This process continues until a predefined stopping criterion is reached, and the prediction is typically obtained by averaging the target values within each leaf node. To reduce overfitting and improve predictive performance, ensemble tree methods are commonly adopted, including Random Forest [185], Gradient Boosted Decision Trees (GBDT) [186], Extreme Gradient Boosting (XGBoost) [187], and LightGBM [188]. Tree-based methods have played an important role in TSF competitions [189] and have been widely applied to domains such as traffic forecasting [100], healthcare [101], and energy forecasting [102].

*3) K-Nearest Neighbor Regression:* K-Nearest Neighbor (KNN) regression is a non-parametric method that does not require explicit model training [190]. It predicts the target value of a given sample by measuring its distance from training samples, using metrics such as Euclidean distance or Dynamic Time Warping (DTW), selecting the $K$ nearest neighbors, and averaging their target values. Given a sample $\mathbf{x}$, the prediction $\hat{y}$ is formulated as:

$$\hat{y} = \frac{1}{K} \sum_{i \in N(\mathbf{x})} y_i, \tag{4}$$

where $N(\mathbf{x})$ denotes the set of $K$ nearest neighbors of $\mathbf{x}$, and $y_i$ represents their corresponding target values. In time series forecasting, KNN regression has been used to model repetitive patterns [103] and seasonality [104]. It has also been extended to multivariate time series forecasting settings [105], [106].

### B. Neural Networks and Deep Learning Models

With the rapid advancement of deep learning, numerous neural forecasting models have been proposed, achieving remarkable forecasting accuracy by effectively capturing both temporal and cross-channel dependencies. In this section, we review recent advancements in deep learning-based forecasting approaches, with a focus on mainstream architectures, as illustrated in Fig. 5.

*1) Recurrent Neural Networks:* Recurrent Neural Networks (RNNs) are widely used for modeling sequential data due to their recursive structure, which captures temporal dependencies. The basic RNN formulation is:

$$\begin{aligned} h_t &= \sigma(U h_{t-1} + V x_t + b_h), \\ \hat{y}_t &= W h_t + b_y, \end{aligned} \tag{5}$$

where $x_t$, $y_t$ and $\hat{y}_t$ represent the input, hidden state, and output at time step $t$, respectively. $h_{t-1}$ is the hidden state at the previous time step $t-1$. $U$, $V$, $W$ are weight matrices, $b_h$, $b_y$ are biases, and $\sigma$ is a non-linear activation function.

RNNs are widely used in TSF [191]. DeepAR [19], MQRNN [192], and DF-Model [110] model predictive uncertainty. DeepAR learns shared temporal patterns and seasonality to generate future distributions. MQRNN uses an LSTM encoder and a dual-MLP decoder for multi-quantile prediction, while DF-Model combines RNNs with Gaussian processes to capture global and series-specific dynamics. For point forecasting, DA-RNN [108] and MH-TAL [109] use attention mechanisms to enhance temporal context modeling. DA-RNN introduces input and temporal attention in an encoder-decoder structure, whereas MH-TAL applies multi-head attention to align historical and future features. Other variants include CRU [25] for irregular time intervals and SegRNN [112] for segment-wise recurrent modeling. Hybrid architectures further combine RNNs with other techniques. LSTNet [107] integrates CNNs and RNNs to capture short- and long-term dependencies, while ES-LSTM [111] combines exponential smoothing with LSTM for hierarchical forecasting. Overall, RNN-based models are flexible for variable-length sequences and sequential dependency modeling. However, they struggle with long-term dependencies due to vanishing or exploding gradients [193], [194], and their sequential computation limits parallelization. Recent models such as xLSTM [195] offer promising directions, but remain underexplored in TSF.

*2) Convolutional Neural Networks:* Convolutional Neural Networks (CNNs), originally established in computer vision and NLP, have also shown strong performance in time series analysis. The core convolution operation is defined as:

$$Y(t) = \sum_{c=1}^{C} \sum_{k=0}^{K-1} X_c(t+k) \cdot h_c(k), \tag{6}$$

where $X$ is the input time series sequence of length $L$ with $C$ channels, $h$ is the convolutional kernel of size $K$, and $Y$ is the output of the convolution operation at time $t$ for a single output channel. CNNs offer three main advantages for time series

modeling: local connectivity, weight sharing, and translation invariance [196]. These properties enable efficient learning of localized patterns that may shift over time, while reducing parameter counts and improving computational efficiency. Given these strengths, CNNs are considered strong candidates for sequence modeling [197].

The Temporal Convolutional Network (TCN) [197] introduces dilated convolutions to enlarge the receptive field without increasing network depth. Later models extend this idea along different directions. MLCNN [113] uses deep multi-layer CNNs to learn high-level representations, while DSANet [198] and MICN [114] employ multi-scale kernels to capture local and global patterns. SCINet [115] organizes convolutions in a binary tree with downsampling to alleviate the limited lower-layer receptive field of TCN. FTMixer [116] further combines time- and frequency-domain convolutions for joint feature extraction. Recent studies also explore generic CNN-based frameworks. TimesNet [117] folds sequences according to dominant periods and applies 2D convolutions. ModernTCN [59] adopts depthwise separable convolutions and reparameterization for efficient large-kernel modeling. TS2Vec [118], built on TCN, uses contrastive learning to enhance timestamp-level representations, while ConvTimeNet [62] employs multi-scale large kernels to capture deep and global features. Overall, CNN-based models effectively extract local patterns for TSF, but may be limited in modeling long-range dependencies.

*3) Attention-Based Neural Networks:* The Transformer architecture [199] has achieved remarkable success across various domains in deep learning, due to its powerful attention mechanism. The core attention operation is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathbf{T}}}{\sqrt{d}}\right)\mathbf{V}, \tag{7}$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are the query, key, and value vectors with dimension $d$ respectively. Based on this, the Transformer is stacked by multiple blocks which function as:

$$\begin{aligned}\mathbf{H}'_l &= \text{LayerNorm}(\text{SelfAttn}(\mathbf{H}_l) + \mathbf{H}_l),\\ \mathbf{H}_{l+1} &= \text{LayerNorm}(\text{FFN}(\mathbf{H}'_l) + \mathbf{H}'_l),\end{aligned} \tag{8}$$

where $\text{SelfAttn}$ denotes attention with query, key, and value derived from the same input. Besides, $\mathbf{H}_l$ is the input of $l-$th block, $\text{FFN}$ is the feed-forward network and $\text{LayerNorm}$ refers to the layer normalization operation. Owing to its iterative design and non-sequential computation, the Transformer excels at capturing long-range dependencies. Numerous variants have since been proposed to tailor it for TSF [13].

Early Transformer-based models usually treat each multivariate observation at a time step as a single token [29]. However, due to the noisy and continuous nature of time series, individual points often lack rich semantic information. To address this, PatchTST [57] groups adjacent time steps into patches to improve locality and representation quality, while iTransformer [122] constructs variate-level tokens to better model inter-channel dependencies. Meanwhile, standard self-attention has $O(L^2)$ complexity, which is costly for long sequences. Efficient variants have therefore been developed, including LogTrans [119] with sparse convolutional self-attention, Autoformer [120] with an Auto-Correlation mechanism of $O(L \log L)$ complexity, and FEDformer [121] with frequency-domain attention. Recent models further address heterogeneous distributions, disrupted correlations, and non-stationarity, such as DUET [123], Non-stationary Transformer [200], and DTAF [124].

Regarding architectural design, early Transformer variants mainly used encoder-decoder structures [120], [200], where the encoder extracts temporal features and the decoder predicts future values. Recent studies suggest that complex decoders may be unnecessary, as encoder-only models with linear output layers achieve improved accuracy and efficiency [57], [201]. Meanwhile, auto-regressive decoder-only Transformers are increasingly used in time series foundation models for modeling and generating sequences of arbitrary lengths [44], [138]. Overall, Transformer-based models are effective for long-range dependency modeling and flexible forecasting, but their high data requirements may cause overfitting on domain-specific or small-scale datasets.

*4) Multi-Layer Perceptrons:* MLP-based models efficiently capture linear correlations through interpretable linear transformations. Architecturally, N-BEATS [125] is a representative model using a deep residual MLP stack for time series forecasting. Building on it, NHits [42] uses hierarchical interpolation and multi-rate sampling to reduce complexity and capture volatility. Koopa [126] separates time-variant and invariant components with Fourier filters and models their dynamics through an MLP-based Koopman predictor. TimeMixer [129] further captures multiscale patterns with a fully MLP-based architecture. Recent studies also explore lightweight MLP-based forecasting for improved efficiency. DLinear [32] shows that simple linear transformations with decomposition can outperform complex architectures. FITS [127] models amplitude and phase using complex-valued networks with only 10 k parameters, while SparseTSF [128] reduces the model to 1 k parameters through cross-period sparse forecasting. SRSNet [130] combines selective patching with an MLP head, showing the benefits of flexible patch selection and reassembly. Overall, MLP-based models are efficient, though large-scale scalability remains open.

### C. Emerging Network Architectures

*1) Deep State Space Models:* State Space Models (SSMs) represent dynamic systems through linear or nonlinear equations that describe state evolution over time, and they have emerged as promising alternatives for sequence modeling. Rangapuram et al. [131] propose a deep state space model for probabilistic time series forecasting by parametrizing a per-time-series linear state space model with a jointly learned recurrent neural network, combining the data efficiency and interpretability of SSMs with the representation learning capability of deep learning. SpaceTime [132] further introduces an SSM parameterization tailored to autoregressive time series processes. Moreover, with the proposal of advanced SSM structures such as Mamba [202], many works have attempted to apply SSMs to time series forecasting and have achieved promising results [133], [134].

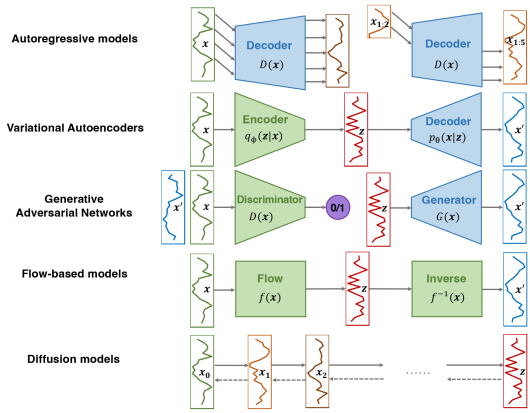



Fig. 6. Representative generative forecasting models for uncertainty modeling, future trajectory synthesis, and conditional predictive distribution estimation.

*2) Kolmogorov-Arnold Networks:* Kolmogorov-Arnold Networks (KANs) [135] are recently proposed architectures that may serve as alternatives to MLPs by applying learnable activation functions on connections between nodes. Some pioneering works have begun to validate their effectiveness in time series tasks, including general time series analysis [136] and forecasting [137].

### D. Deep Time Series Generative Models

Generative models have become increasingly valuable in time series forecasting because they can model underlying data distributions and generate diverse future samples. Unlike discriminative models that estimate $P(Y|X)$ for prediction, generative models learn the joint distribution $P(X, Y)$, enabling both forecasting and realistic data synthesis. Fig. 6 provides a schematic overview of common generative models.

*1) Autoregressive Models:* Autoregression is a fundamental concept in time series forecasting. However, due to the non-stationarity of time series and the requirement for high-precision prediction [12], autoregressive methods often suffer from error accumulation and remain relatively underexplored in this domain. The classical ARIMA model [28] mitigates this issue by combining autoregressive and moving average components with differencing. Recently, the success of large pre-trained language models has revived interest in autoregressive approaches for time series [73], [138], benefiting from their generalization and flexibility. Some models have even been extended to support arbitrary input-output mappings.

*2) Variational Autoencoders (VAEs):* VAEs are probabilistic generative models that encode inputs into a latent space and decode them to reconstruct the original data [203]. This enables VAEs to model complex distributions and generate new samples, making them suitable for time series tasks involving uncertainty and structured latent representations. TimeVAE [20] proposes a VAE architecture for multivariate time series generation, emphasizing interpretability and efficiency. It incorporates domain knowledge (e.g., trends and seasonality) to produce interpretable outputs and performs well on similarity and next-step prediction tasks. Besides, HyVAE [139] enhances VAEs through hybrid variational inference, improving the modeling of local patterns and temporal dynamics for time series forecasting.

*3) Generative Adversarial Networks (GANs):* GANs consist of a generator that produces synthetic data and a discriminator that distinguishes it from real data [204]. This adversarial setup enables GANs to generate realistic samples, although modeling sequential dependencies in time series remains a key challenge. TimeGAN [140] addresses this issue by incorporating recurrent structures into both the generator and the discriminator, thereby capturing temporal dynamics while preserving statistical properties. It supports tasks such as simulation, augmentation, and anomaly detection. Recent extensions include a context-aware GAN for traffic forecasting on new roads [141], which integrates external factors such as weather and weekdays, and CurbGAN [142], which employs conditional GANs with dynamic convolutions and self-attention to model traffic patterns under varying demand.

*4) Flow-Based Models:* Normalizing Flow-based models transform a simple base distribution into a complex target distribution via invertible and differentiable mappings [205]. This enables exact likelihood estimation and is effective for modeling high-dimensional data with complex dependencies. MAF [21] leverages flows to model multivariate sequences by conditioning on past observations, capturing rich temporal structures. Recent developments have further extended this technique in various ways. CFMTS [143] enhances neural ODE training by regressing conditional probability paths, improving long-horizon performance. TFM [144] proposes a simulation-free method for training Neural SDEs, offering better stability and scalability, especially in clinical settings. FM-TS [145] introduces a Flow Matching-based framework for efficient and effective time series generation, outperforming diffusion models in both unconditional and conditional tasks.

*5) Diffusion Models:* Denoising Diffusion Probabilistic Models (DDPMs) have become prominent in time series forecasting for capturing complex temporal patterns and generating realistic samples. They use a diffusion-denoising process: noise is added during forward diffusion and removed during reverse diffusion, enabling learning of rich data distributions. Early models like TimeGrad [146] applied autoregressive denoising with Langevin sampling. TSDiff [147] improved short-term accuracy via self-guidance, while ScoreGrad [148] employed SDEs for continuous-time forecasting of irregular data. Conditional models such as TimeDiff [149] and DiffLoad [206] leveraged external signals for improved forecasting, with applications to power load, ICU, and ECG data [207], [208]. Recent advances include Latent Diffusion Models (LDMs), like Latent Diffusion Transformers (LDT) [209], which enhance scalability and performance in tasks like precipitation forecasting. Functional-space diffusion models (e.g., DSPD, CSPD [49]) support anomaly detection and interpolation. FDF [150] integrates linear and conditional components to better capture trends and seasonality.

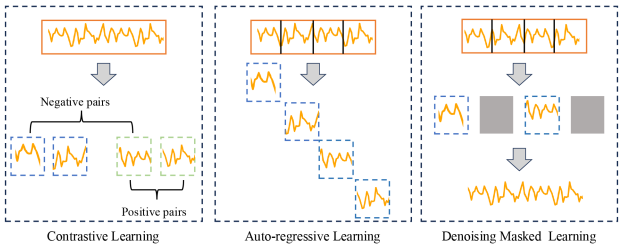



Fig. 7. Illustration of major self-supervised pre-training strategies for time series.

## VII. Transfer Learning Methods

In this section, we introduce the transfer learning techniques for TSF, including self-supervised pre-training, time series foundation models, and LLM-based methods.

### A. Self-Supervised Pre-Training Methods

Self-supervised learning alleviates the dependence on labeled datasets by enabling models to learn transferable representations through pre-training on unlabeled data. Prominent techniques include contrastive learning, denoising masked autoencoders, and autoregressive pre-training models. The proposed taxonomy is illustrated in Fig. 7.

*1) Contrastive Learning Model:* The fundamental principle of contrastive learning is to allow the model to differentiate between positive and negative samples. This is achieved by drawing positive samples closer together and pushing negative samples further apart, thereby enabling the model to develop a highly discriminative representation. For time series, contrastive learning frequently employs data augmentation techniques such as jittering, temporal cropping, and masking to generate varied perspectives of the samples, thereby enhancing the model's generalization across diverse scenarios. TS2Vec [118] serves as a foundational contrastive learning framework for time series, introducing a universal approach to time series representation. Building upon this groundwork, TNC [151] and TS-TCC [152] further advance unsupervised representation techniques, refining the model's capacity for feature extraction in time series. TF-C [153] extends this framework by incorporating frequency domain information and employing contrastive learning to enforce consistency between time and frequency domains, thereby enabling the model to capture stable and generalizable representations across diverse datasets.

*2) Denoising Masked Autoencoder Model:* The masked autoencoder model strategically masks portions of the input data, allowing the model to learn robust data representations through a self-supervised learning objective by reconstructing the masked segments. In the raw data space, TST [154] establishes the masked autoencoder framework, while PatchTST [57] further advances it by introducing channel independence and partitioning time series into patches, thereby enhancing model flexibility in time series representation. Within the discrete latent space, TimeMAE [155] and CrossTimeNet [69] project time series into a discrete representation space, unifying the input format and facilitating cross-domain self-supervised pre-training across heterogeneous datasets. In the manifold space, SimMTM [156] reconstructs masked time points through weighted aggregation of adjacent time slices.

*3) Auto-Regressive Pre-Training Model:* Auto-regressive models capture time-dependent characteristics by leveraging historical data to predict values at future time points. In time series, auto-regressive pre-training models progressively learn sequence dynamics and temporal dependencies by modeling historical patterns. TimeDART [157] uses time series blocks as fundamental modeling units, employing a self-attention-based Transformer encoder to establish inter-block dependencies and incorporating diffusion and denoising mechanisms to capture fine-grained features within blocks, thereby effectively applying diffusion models to time series. GPHT [44] constructs a hybrid dataset for pre-training under the channel independence assumption, significantly expanding the data scale. Additionally, it employs an autoregressive prediction method to model the temporal dependencies of the output sequence without a custom prediction head, enabling adaptation to various prediction lengths and providing new insights into self-supervised pre-training. Timer [138] demonstrates strong scalability and generalization by unifying heterogeneous time series into a single sequence format (S3) and utilizing a GPT-style decoder structure.

### B. Time Series Foundation Models

We distinguish LM-based forecasting methods from time-series foundation models. LM-based methods explicitly use pretrained language models as the forecasting backbone, semantic interface, or reasoning engine, often through prompting, fine-tuning, reprogramming, or textual context integration. In contrast, time-series foundation models are pretrained on large-scale time series corpora with objectives centered on temporal data. Therefore, not all time-series foundation models are LM-based, and not all LM-based methods are time-series foundation models. Recently, time-series foundation models have attracted increasing attention due to their cross-domain transferable representations. Their architectures generally follow two paradigms: encoder-based structures for universal temporal feature extraction and decoder-based structures for autoregressive generation.

The encoder-based approaches include MOMENT [158], which integrates multi-domain data from transportation and healthcare to establish a multi-task pretraining system that supports classification, forecasting, and anomaly detection. Following similar principles, Moirai [45] introduces attention mechanisms for arbitrary variates and achieves generalizable forecasting through masked pretraining on the LOTSA dataset, demonstrating effectiveness in out-of-distribution scenarios such as electricity load forecasting. In contrast, generative foundation models exhibit distinct design philosophies: TimesFM [159] combines patch techniques with decoder structures and frequency-specific tokenization to enable zero-shot generalization, while Chronos [74] discretizes continuous time series into token buckets and leverages synthetic data with

TABLE IV
COMPARISON AND PRACTICAL GUIDANCE OF MAJOR TIME SERIES FORECASTING PARADIGMS

| Paradigm | Modeling idea | Preferable conditions | Key limitations |
|---|---|---|---|
| **Statistical modeling** | Handcrafting parametric assumptions. | Small-scale data with regular sampling or autocorrelation. | Limited ability to model nonlinear dependencies. |
| **Machine learning** | Manually extracting informative features from raw time series. | Medium-scale data where domain features are available. | Performance depends heavily on feature engineering and may not capture complex sequential dependencies. |
| **Deep learning** | Learning representations from time series for direct prediction. | Large-scale data with nonlinear temporal dependencies. | Sensitive to preprocessing, distribution shift, data scale, and hyperparameter settings. |
| **Foundation-model-based forecasting** | Learning transferable representations through large-scale pretraining. | Cross-domain, few-shot, zero-shot, or low-resource forecasting settings. | Key challenges include domain mismatch, heterogeneous semantics, unclear scaling, and immature evaluation. |
| **LLM-based forecasting** | Aligning LLMs with time series through adaptation or reasoning. | Forecasting tasks involving contextual features. | High computational cost, unstable reasoning, and potential hallucination. |

cross-entropy training for zero-shot probabilistic forecasting. Timer [138] further unifies diverse downstream tasks within one framework through decoder-only architectures and autoregressive generative training strategies.

Despite architectural variations, these models share fundamental principles of simplicity and generality, avoiding overly specialized designs. Current foundation models focus primarily on constructing large-scale temporal datasets, developing cross-domain modeling techniques, and establishing unified frameworks for multivariate analysis. Notably, the data scale surpasses previous end-to-end methods by orders of magnitude, as exemplified by 27B time steps in LOTSA. Furthermore, synthetic time series are emerging as a critical pretraining resource, with synthetic data constituting 20% of TimesFM's pretraining corpus.

*Key takeaway:* Foundation models show promise for zero- and few-shot forecasting, but their effectiveness depends on pretraining data quality and task-objective alignment, motivating benchmarks such as TSFM-Bench [210]. Table IV summarizes the assumptions, applicable conditions, and limitations of major forecasting paradigms. Taxonomically, foundation models represent an emerging large-scale pretraining and transfer paradigm, while remaining closely related to data-driven forecasting through representation learning.

### C. LM-Based Models

LM-based methods leverage LLMs for forecasting by exploiting their semantic representation and reasoning capabilities [211]. Compared with traditional predictors, they can further integrate contextual textual information, such as individual characteristics or sampling backgrounds, with numerical features through prompting or quantization techniques [166], [212]. LM-based predictors can be broadly divided into tuning-based and tuning-free approaches.

*1) Tuning-Based:* Tuning-based methods adapt LLMs to specific forecasting tasks by updating model parameters using time series. This typically involves additional training to adjust weights for better alignment with temporal patterns. From the forecasting paradigm perspective, models apply autoregressive generation, simulating left-to-right sequence construction [138], [157], while FPT [160] avoids error accumulation by generating the full prediction in one step. In terms of training strategies, CrossTimeNet [69] introduces a self-supervised pretraining stage to enhance time series representation learning, and TEMPO [166] incorporates textual inputs to support prediction. Regarding parameter updates, LLM4TS [165] adopts LoRA [213] for low-rank fine-tuning of key components, whereas others [69], [160] update partial or full base model parameters. For input processing, Chronos [74] and UniTime [161] embed numerical time series directly, while TimeLLM [162] fuses temporal and textual features to enrich input. These diverse strategies contribute to the performance of tuning-based LLM forecasters.

*2) Tuning-Free:* Tuning-free methods directly apply pretrained LLMs to time series forecasting without fine-tuning, enabling rapid deployment with lower computational cost. LLM-Time [164] incorporates sampling background into prompts, while LSTPrompt [214] introduces a TimeBreak mechanism that pauses after several predictions to mimic human reasoning. In clinical settings, Zhu et al. [215] use Electronic Health Records and in-context learning aligned with real-world workflows. Retrieval-based methods, such as TimeRAF [216] and TimeRAG [217], construct prompts by retrieving similar time series or encoding neighboring samples in JSON format. Table-Time [218] evaluates Llama-3.1-405b-instruct for time series classification and understanding. From an agent perspective, TESSA [219] uses domain-specific annotation agents to generate descriptive text, improving LLM comprehension and reasoning over time series. However, Merrill et al. [220] show that tuning-free methods may perform only slightly better than random guessing on reasoning tasks, and contextual enhancements bring limited gains. These findings indicate that tuning-free LLM-based TSF remains promising but underdeveloped.

## VIII. TRUSTWORTHY TIME SERIES FORECASTING

As forecasting models are increasingly applied in critical domains such as finance and energy management, the demand for trustworthy TSF continues to grow. We next review advances in trustworthy TSF from three aspects: interpretability, robustness, and privacy preservation.

### A. Interpretability

Trustworthy TSF requires both accurate predictions and interpretability, enabling users to understand and trust the prediction process. Mainstream interpretability research mainly follows two directions: causal discovery and physics-informed neural networks (PINNs). Causal discovery improves forecasting by identifying cause-and-effect relationships among variables [221]. It reveals how variables influence each other over time, enhancing accuracy and interpretability, especially in systems where predictions depend on causal structures [222].

For example, VAR models use Granger causality for prediction [167], while dynamic Bayesian networks capture temporal dependencies and evolving causal structures [168]. Deep models can also incorporate causal inference to improve interpretability and handle complex non-stationary patterns [169]. PINNs provide another promising direction by incorporating physical principles, such as conservation laws and differential equations, into data-driven models [172]. This helps generate predictions consistent with both observations and physical constraints, improving accuracy in TSF [170].

### B. Robustness

Despite significant advancements in recent forecasting models, they remain vulnerable to adversarial attacks, which raises important concerns about their trustworthy deployment in critical applications. Understanding the robustness of these models against malicious attacks and developing effective defense mechanisms is crucial. To address this, Dang-Nhu et al. [173] introduce an effective adversarial attack generation method through Monte Carlo estimation of statistics from the joint distribution of the target series, applied to probabilistic autoregressive forecasting. Liu et al. [174] identify a new attack pattern involving strategic, sparse modifications and propose defense strategies using randomized smoothing techniques and adversarial training. Besides, RDAT [175] employs a reinforcement learning-based adversarial training approach with self-knowledge distillation regularization to enhance the adversarial robustness of spatiotemporal traffic forecasting models. Another notable work, BACKTIME [176], investigates backdoor attacks on TSF tasks.

### C. Privacy

With the increasing demand for time series foundation models, a promising approach is to train using vast amounts of time series from multiple sources. However, this raises significant concerns regarding data privacy. To address this issue, several efforts have focused on integrating the federated learning paradigm, enabling the use of multi-domain data while preventing the exposure of sensitive information. CNFGNN [177] explicitly encodes the graph structure under the constraints of cross-node federated learning to effectively model complex dependencies in spatio-temporal forecasting, ensuring that data privacy is maintained while still capturing intricate relationships across nodes. MetePFL [178] introduces a federated prompt learning mechanism, combined with a graph-based approach, to mitigate the impact of data heterogeneity while preserving data security. Moreover, Time-FFM [179] proposes a prompt adaptation module and a personalized federated training strategy by learning global encoders and local prediction heads to construct a federated foundation model for TSF by leveraging pretrained language models.

## IX. Prospects and Future Directions

In this section, we discuss the future directions of time series forecasting from various perspectives.

### A. Time Series Foundation Models

Inspired by the success of foundation models in vision and natural language processing, developing time series foundation models has become a promising direction. Although recent studies show encouraging progress, several challenges remain. A key direction is to build context-aware foundation models for general TSF. Existing methods often rely on large-scale time series pretraining to improve generalization, but may overlook contextual characteristics such as sampling frequency, noise level, physical meaning, subject-specific patterns, and task requirements. Therefore, effectively integrating multimodal contextual information, such as text, images, and graphs, remains an important challenge. Whether universal foundation models are optimal for TSF remains unclear. Future work should study scaling laws, emergent behaviors, pretraining data composition, and the role of domain-specific priors. It should also clarify when foundation models outperform tuned statistical or discriminative baselines under varying data scales, domain diversity, and covariate availability. Another key issue is how time series tokenization can preserve numerical precision while enabling semantic alignment.

### B. Trustworthy Time Series Forecasting

Most mainstream TSF methods rely on closed-box neural networks, which achieve high accuracy but remain difficult to deploy in sensitive applications such as healthcare and decision-making. A promising direction is to combine explainable AI and causal inference with powerful forecasting models to build interpretable and trustworthy forecasting systems. With multi-source cross-domain data increasingly used, privacy and security have become key concerns. Although federated learning is widely adopted, it still faces challenges in protecting client-side time series and resisting poisoning attacks from malicious clients. Another open issue is evaluating probabilistic and generative forecasters beyond point accuracy. In decision-critical scenarios, calibration, tail risk, interval coverage, and downstream utility may matter more than marginal MAE or MSE gains. For deployment, distribution shift and covariate availability are especially urgent, while long-horizon stability, uncertainty estimation, and interpretability remain critical in high-stakes applications.

### C. Emerging Modeling Paradigms

TSF has always been an open research field, encouraging the validation of diverse techniques, including different neural network architectures and model training paradigms. These methods, in turn, provide valuable insights that propel the further development of the entire field. Therefore, combining emerging technologies from the broader machine learning domain with the unique characteristics of time series is a promising future direction for research. For example, exploring how to integrate automated machine learning methods to adjust network architectures, balance computational costs with forecasting accuracy, and how to incorporate physical knowledge of the time series through Physics-Informed Neural Networks (PINNs) are potential areas for future investigation. This direction also

raises concrete modeling questions: how can new architectures combine temporal, spectral, graph, and textual signals without introducing excessive complexity, and how can automated design methods remain interpretable enough for domain experts to trust?

### D. Comprehensive Benchmark Evaluation

Beyond model design, performance evaluation and benchmark construction are important future directions for TSF. Existing benchmarks often cover limited data distributions, insufficiently distinguish task difficulty, and rely on overly simple metrics, making it hard to assess models comprehensively and potentially encouraging hyperparameter-driven performance chasing. Thus, more general evaluation datasets and diverse metrics are needed to support healthier and more balanced progress in the field. Future benchmarks should separately attribute gains to model architecture, pretraining data, covariate access, normalization, and forecasting strategy. They should also cover missing or partially observed future covariates, distribution shifts, and heterogeneous domains, ensuring that improvements reflect realistic difficulty rather than information leakage or favorable splits. Standardized benchmarks should report consistent splits, horizons, preprocessing, tuning budgets, and covariate-availability assumptions across model families.

## X. Conclusion

In this survey, we present a holistic and structured review of recent advances in time series forecasting, covering foundational concepts, methodological developments, and critical challenges. We provide a unified perspective that connects classical statistical modeling with modern deep learning paradigms, highlighting how data availability, computational resources, and algorithmic progress are reshaping the field. By examining key challenges, including non-stationarity, uncertainty quantification, high dimensionality, and interpretability, we illustrate the complexity of modern forecasting tasks. We further identify emerging directions, such as generative forecasting, explainable AI, and integrative frameworks that combine domain knowledge with data-driven modeling. Overall, this survey consolidates the state of the art and outlines promising research opportunities, providing a coherent reference for researchers and practitioners in TSF.

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